



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO2_AT1

Case Study Title: Smart Waste Collection System

(SDG 11: Sustainable Cities and Communities)

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1. Scenario Understanding and Problem Analysis

Business Domain

Smart City / Municipal Waste Management

Problem Statement

The current waste collection process follows fixed schedules without checking whether waste bins are full or empty. As a result, some bins overflow before collection, causing unhygienic conditions and environmental pollution, while other bins are collected even when nearly empty, leading to unnecessary fuel consumption, increased operational costs, and inefficient use of manpower.

Stakeholders

- Municipal Administrator
- Waste Collection Drivers
- Sanitation Workers
- Citizens
- City Management
- IoT Sensor Monitoring Team
- System Administrator

Existing Challenges

- Fixed waste collection schedules
- Overflowing waste bins
- Unnecessary vehicle trips
- High fuel consumption
- Poor route planning
- Increased operational costs
- Lack of real-time monitoring
- Citizen complaints due to delayed waste collection

System Objectives

- Monitor bin fill levels in real time.
- Optimize garbage collection routes.
- Reduce fuel and operational costs.
- Prevent overflowing bins.
- Improve waste collection efficiency.

- Provide reports and analytics for decision-making.

2. Functional Requirements

FR No	Functional Requirement
FR1	User login and authentication.
FR2	Register and manage smart waste bins.
FR3	Monitor real-time waste bin fill levels.
FR4	Generate alerts when bins reach threshold levels.
FR5	Optimize collection routes automatically.
FR6	Assign routes to collection vehicles and drivers.
FR7	Track vehicle location using GPS.
FR8	Record waste collection status.
FR9	Citizens can report overflowing bins.
FR10	Generate daily, weekly, and monthly reports.
FR11	Maintain collection history.
FR12	Send notifications to administrators and drivers.
FR13	Manage user roles and permissions.
FR14	Dashboard for monitoring overall system performance.

3. Non-Functional Requirements

Performance:

- The system should respond within **3 seconds**.
- Support monitoring of thousands of smart bins simultaneously.

Security:

- Secure login with authentication.
- Role-based access control.
- Encrypted data transmission.

Reliability:

- 99.9% system availability.
- Automatic data backup.

Usability:

- Simple and user-friendly interface.

- Easy navigation for administrators and drivers.

Scalability:

- Easily expandable to multiple cities.
- Support additional waste bins and vehicles.

Maintainability:

- Modular software architecture.
- Easy software updates and bug fixing.

Availability:

- 24×7 system availability.
- Cloud-based access.

4. Actors and Use Cases**Primary Actors**

- Municipal Administrator
- Driver
- Citizen

Secondary Actors

- IoT Sensor
- GPS Service
- Database Server

Major Use Cases**Municipal Administrator**

- Login
- Add smart bins
- View dashboard
- Monitor waste levels
- Assign collection routes
- Manage drivers
- Generate reports
- View complaints

Driver

- Login
- View assigned route
- Navigate using GPS
- Update collection status

Citizen

- Register/Login
- Report overflowing bins
- View complaint status

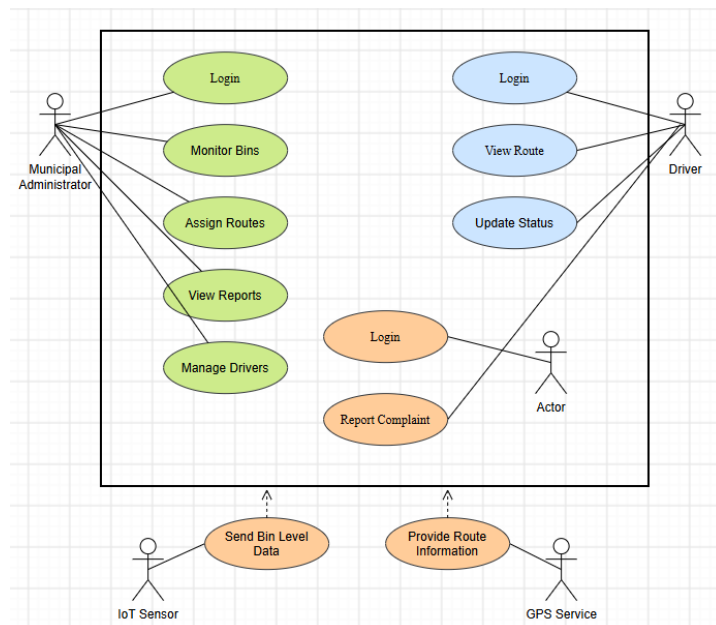
IoT Sensor

- Send bin level data
- Trigger alerts

GPS Service

- Provide vehicle location
- Route optimization

5. Use Case Diagram (Text Representation)



6. Software Requirement Specification (SRS)

6.1 Introduction

The Smart Waste Collection System is designed to improve municipal waste management using IoT-enabled smart bins, GPS-based route optimization, and real-time monitoring. The

system helps reduce operational costs, improve sanitation, and enhance the efficiency of waste collection.

6.2 Problem Description

Traditional waste collection follows fixed schedules without considering actual bin status. This results in overflowing bins, unnecessary fuel usage, increased costs, inefficient manpower utilization, and poor citizen satisfaction.

6.3 Scope of the System

The system will:

- Monitor waste bin levels.
- Optimize collection routes.
- Track collection vehicles.
- Manage complaints.
- Generate analytical reports.
- Improve operational efficiency.

6.4 Functional Requirements

- User authentication
- Smart bin monitoring
- Route optimization
- Vehicle tracking
- Waste collection updates
- Complaint management
- Notification services

6.5 Non-Functional Requirements

- High performance
- Secure communication
- Reliable operation
- User-friendly interface
- Cloud scalability

6.6 Use Case Descriptions

Use Case 1 – Monitor Waste Bin

Actor: Municipal Administrator

Precondition: User is logged in.

Flow:

1. Administrator opens dashboard.
2. System displays bin status.
3. Administrator identifies full bins.
4. Route optimization is initiated.

Postcondition:

Collection route is prepared.

Use Case 2 – Update Collection Status

Actor: Driver

Precondition: Driver is logged in.

Flow:

1. Driver views assigned route.
2. Collects waste.
3. Updates collection status.
4. System records completion.

Postcondition:

Collection history is updated.

Use Case 3 – Report Overflowing Bin

Actor: Citizen

Precondition: Citizen is registered.

Flow:

1. Citizen logs in.
2. Selects complaint option.
3. Uploads complaint.
4. System notifies administrator.

Postcondition:

Complaint is stored for action.

6.7 Data Requirements

Inputs

- Bin fill level

- GPS location
- Driver details
- Vehicle details
- Complaint details
- User credentials

Outputs

- Optimized routes
- Bin status
- Reports
- Notifications
- Collection history
- Complaint status

Database Entities

- Users
- Smart Bins
- Drivers
- Vehicles
- Collection Schedule
- Collection History
- Complaints
- Reports

6.8 External Interface Requirements

User Interface

- Web dashboard for administrators
- Mobile application for drivers
- Citizen complaint portal

Hardware Interface

- Smart waste bins
- IoT sensors
- GPS devices

Software Interface

- Cloud database
- GPS API
- Notification service
- Authentication module

Communication Interface

- Internet
- Wi-Fi
- 4G/5G network

6.9 System Features

- Real-time bin monitoring
- GPS route optimization
- Driver management
- Vehicle tracking
- Complaint management
- Notification system

7. Assumptions and Constraints

Assumptions

- All smart bins are equipped with IoT sensors.
- GPS devices are installed in collection vehicles.
- Internet connectivity is available.
- Municipal staff are trained to use the system.
- Cloud infrastructure is available.

Constraints

Technical

- Depends on IoT sensor accuracy.
- Requires stable internet connectivity.

Operational

- Drivers must follow optimized routes.
- Regular maintenance of sensors is required.

Economic

- Initial cost for IoT devices and GPS installation.
- Ongoing cloud hosting and maintenance costs.

Legal

- Compliance with municipal regulations.
- Protection of user data and privacy.

Time

- Project should be completed within the planned development schedule.

8. Requirement Validation

Completeness

- Functional and non-functional requirements are identified.
- All stakeholders are considered.
- Major use cases are included.

Consistency

- Requirements are clear and non-conflicting.
- Functional requirements support system objectives.

Ambiguity Check

- Requirements are specific and measurable.
- Roles and responsibilities are clearly defined.

Missing Requirements Check

- Notification system included.
- Complaint handling included.
- Route optimization included.
- Reporting module included.

Suggested Improvements

- Integrate AI-based prediction for waste generation.
- Add multilingual support.
- Enable offline mode for drivers.
- Include carbon emission tracking.
- Provide citizen mobile notifications.

Conclusion:

The proposed Smart Waste Collection System provides an efficient software solution for municipal waste management by combining IoT-based monitoring, GPS route optimization, real-time tracking, and analytics. It addresses the existing challenges of fixed collection schedules, reduces operational costs, improves sanitation, and supports SDG 11: Sustainable Cities and Communities through smarter, more sustainable urban waste management.